

NORMALIZATION

Definition:

- ➔ Normalization is the process of decomposition of the table into smaller tables in order to removing the redundancy of the data as well as different types of anomalies such as insert, delete and update anomalies.
- ➔ We can achieve the normalization concepts by implementing different types of **normal forms**.
- ➔ Normal form is a rule that satisfied the given table i.e, we can design a table in such a way satisfied the given normal form.

Normal Forms:

- ➔ Every Normal Form consisting of a rule and we need to apply that rule on the given tables.
- ➔ According to normalization we have six different normal forms.
 - 1) 1NF
 - 2) 2NF
 - 3) 3NF
 - 4) BCNF or 3.5NF
 - 5) 4NF
 - 6) 5NF

1st Normal Form:

- ➔ A table is said to be 1nf it should be satisfied two given conditions.
 - 1) Every column name in the table must be unique.
 - 2) Every column value must be atomic (single) value.

Example Relation:

empid	empname	empsal	empadd	empphno

111	smith	10000	ong	12345
				86276
222	jhon	20000	hyd	78396
				37829

- ➔ The above table is not in 1NF, empphno column has multiple values. Hence it violates 1 NF rule.
- ➔ We need to apply the 1 NF rule on above table, the resultant table as shown below.

empid	empname	empsal	empadd	empphno

111	smith	10000	ong	12345
111	smith	10000	ong	86276
222	jhon	20000	hyd	78396
222	jhon	20000	hyd	37829

2nd Normal Form:

- ➔ A table is said to be 2NF it must and should satisfied the below two conditions
 - 1) The table must be in 1NF
 - 2) Table should not have partial functional dependency.

Partial Functional Dependency:

- ➔ When a non-key is depends upon the part of prime key attributes is said to be partial functional dependency.

Example: ABC->D

Hear attribute D is a non-key where as attributes A,B and C are the key attributes.

If attribute D depends on subset of ABC then partial dependency is exist among them.

AB -> D or BC -> D or B -> D etc...

- ➔ According to 2NF the table should not have partial functional dependency.
- ➔ Lets take an example table has partial functional dependency

STUDENT TABLE

SID	SNAME	SADD	SUBJID	SUBJNAME	SUBJMARKS
1	SMITH	BANG	101	JAVA	90
2	JHON	HYD	101	JAVA	80
1	SMITH	BANG	102	PYTHON	60
2	JHON	HYD	102	PYTHON	75

Note: The above table is not in 2NF, reason is given below

SID, SUBJID -> SNAME, SUBJNAME, SUBJMARKS

Where SUBJNAME & SUBJMARKS depends on the SUBJID, where SUBJNAME & SUBJMARKS part of the SID & SUBJID so partial dependency is exist in this table hence it does not meet 2NF rule.

➔ We need to decompose the above table into two small tables with the help of the primary key and foreign key

SUBJECT TABLE

SUBJID IS PRIMARY KEY

SUBJID	SUBJNAME	SUBJMARKS
111	C	100
222	PYTHON	100
333	JAVA	100

STUDENT TABLE

SUBJID IS FOREIGN KEY

SID	SNAME	SADD	SUBJID
101	SMITH	BANG	111
102	JHON	HYD	222
101	SMITH	BANG	222
102	JHON	HYD	111
103	MIKE	PUNE	333

sid -> sname, sadd (No partial functional dependency)

From the above tables foreign key (subjid from student table) refer to the primary key (subjid of subject table).

➔ It is observed that no partial functional dependency is exist in these tables

➔ So partial functional dependency was removed hence we can say these two tables in 2 NF.

3rd Normal Form:

➔ A table is said to be a 3rd Normal Form table it must and should satisfied the below two conditions.

- 1) Table must be in 2nd Normal Form and
- 2) It should not have transitive functional dependency.

Transitive Functional Dependency:

Def: when a non-key is depends on another non-key in a given table then we can say that functional dependency is transitive functional dependency.

Example:

In a given relation, the functional dependency $A \rightarrow BC$ is said to be transitive functional dependency when B is dependence on the C attribute or C dependence on B attribute (both B and C are here non-keys)

- ➔ According to 3rd Normal form you should not bring a table that has transitive functional dependency
- ➔ Let's take the below table having the transitive functional dependency

STUDENT TABLE

SID	SNAME	EXAMNAME	TOTAL_MARKS
1	SMITH	INTERNAL	40
1	SMITH	EXTERNAL	60
2	JHON	INTERNAL	40
2	JHON	EXTERNAL	60
3	MIKE	INTERNAL	40
3	MIKE	EXTERNAL	60

- ➔ Observe the above table it has transitive functional dependency
- Here,

SID -> SNAME, EXAMNAME, TOTAL_MARKS

SID can determine SNAME, EXAMNAME, and TOTAL_MARKS uniquely where as TOTAL_MARKS dependence on the EXAMNAME (both TOTAL_MARKS & EXAMNAME NON-KEYS) hence it violate 3rd Normal Form rule.

- ➔ So, we need divide this table into two smaller tables in order to removing the transitive functional dependency.
- ➔ Here table1 is exam table having exam_name is primary key attribute and table 2 is student table having exam_name is foreign key attribute.

EXAM TABLE

EXAM_NAME -> PRIMARY KEY

EXAM_NAME	TOTAL_MARKS
INTERNAL	40
EXTERNAL	60
END EXAMS	100

TABLE 1

STUDENT TABLE

EXAM_NAME -> FOREIGN KEY

SID	SNAME	EXAMNAME
1	SMITH	INTERNAL
1	SMITH	EXTERNAL
2	JHON	INTERNAL
2	JHON	EXTERNAL
3	MIKE	INTERNAL
3	MIKE	EXTERNAL

TABLE 2

- ➔ Here SID -> SNAME, EXAM_NAME
- ➔ Using SID we can determine SNAME and EXAM_NAME uniquely where SNAME and EXAM_NAME both are independent to each hence no transitive functional dependency is exist
- ➔ We can say above tables are in 3rd Normal Form.
- ➔ From the above tables foreign key (EXAM_NAME from STUDENT table) refer to the primary key attribute (EXAM_NAME of EXAM table).

BCNF (BOYCE CODD NORMAL FORM) / 3.5 NF:

➔ A table is said to be BCNF table it must and should satisfied the below two conditions

- 1) Table must be in 3NF
- 2) For every $A \rightarrow B$, A should be a super key

In details,

When non-key determines the key attribute is called the trivial functional dependency that means in every functional dependency LHS attributes should not be non-key.

- ➔ According to BCNF in every functional dependency LHS attribute must be a super key (key attribute)
- ➔ Lets take an example table having the trivial functional dependency

STUDENT TABLE

SID	SUBJNAME	PROFESSOR
101	JAVA	PROF.JAVA
101	CPP	PROF.CPP
102	JAVA	PROF.JAVA2
102	C#	PROF.C#
103	JAVA	PROF.JAVA
103	PYTHON	PROF.PYTHON

- ➔ It is observed that, the above relation does not meet the BCNF rule,
Reason,
From the above table there are two functional dependencies are existed,
FD1: $SID \rightarrow SUBJNAME, PROFESSOR$
FD2: $PROFESSOR \rightarrow SUBJNAME$
- ➔ Using SID we can determine the SUBJNAME and PROFESSOR data uniquely where as PROFESSOR can lead the SUBJECTS
- ➔ So in this $PROFESSOR \rightarrow SUBJNAME$ functional dependency PROFESSOR is not a super key, hence it violate the BCNF rule.

- ➔ So, we need to divide the above table into two smaller tables to make for every functional dependency LHS attribute must be a super key.

PROFESSOR TABLE
PID -> PRIMARY KEY

PID	PROFESSOR	SUBJECT
101	PROF.JAVA	JAVA
102	PROF.PYTHON	PYTHON
103	PROF.C#	C#
104	PROF.JAVA2	JAVA
105	PROF.CPP	CPP

TABLE 1

STUDENT TABLE
PID -> FOREIGN KEY

SID	SNAME	PID
1	SMITH	101
1	SMITH	103
2	SCOTT	104
2	SCOTT	105
3	CLERK	102
3	CLERK	103

TABLE 2

- PID -> SUBJECT** **SID -> SNAME, PID**
- ➔ From the above two relations in every functional dependency LHS attribute is a super key
- i.e, PID -> SUBJECT PID IS A SUPER KEY
- SID -> SNAME, PID SID IS A SUPER KEY

4TH NORMAL FORM:

- ➔ A table is said to be in 4th normal form it should be satisfied two conditions given below.
- 1) It should be in BCNF table
 - 2) It should not have multivalued functional dependency.

Multivalued Functional Dependency:

Def: a functional dependency between A -> B is said to be multivalued functional dependency, when for every single value of attribute A there exist multiple values from B attributes.

i.e, A -> B, C in this functional dependency for every single value of A there exist multiple values from B and C attributes so, A -> B, C functional dependency is also called as multivalued functional dependency.

Causes of occurring multivalued functional dependency is given below

- ➔ When a table has at least three attributes then there is a chance of multivalued dependency exist among them.
- ➔ When a table has at least three attributes and any two attributes are independent to each other then there is a chance of multivalued dependency exist among them.

➔ For every single of attribute there exist multiple values from other attributes then there is a chance of occurring the multivalued dependency.
Let's consider the below table which is not in 4th Normal Form

BIKE_INFO TABLE

BIKE_MODEL_NO	YEAR	COLOR
111	2022	GREEN
111	2023	YELLOW
222	2022	BLUE
222	2023	RED
333	2022	ORANGE
333	2023	PINK

➔ It is observed that the above table has multivalued dependency as shown in below functional dependency

FD: BIKE_MODEL_NO -> YEAR, COLOR

From the above relation a single value of BIKE_MODEL_NO field exists more than one values from YEAR & COLOR attributes and also both YEAR & COLOR attributes independent to each other so multivalued dependency is there in the above table hence it violates 4th normal form rule.

➔ We need to divide the above table into two smaller tables in order to removing the multivalued functional dependency.

BIKE_YEAR TABLE: TABLE1

BIKE_MODEL_NO	YEAR
111	2022
111	2023
222	2022
222	2023
333	2022
333	2023

BIKE_COLOR TABLE: TABLE2

BIKE_MODEL_NO	COLOR
111	GREEN
111	YELLOW
222	BLUE
222	RED
333	ORANGE
333	PINK

FD: BIKE_MODEL_NO -> YEAR

FD: BIKE_MODEL_NO -> COLOR

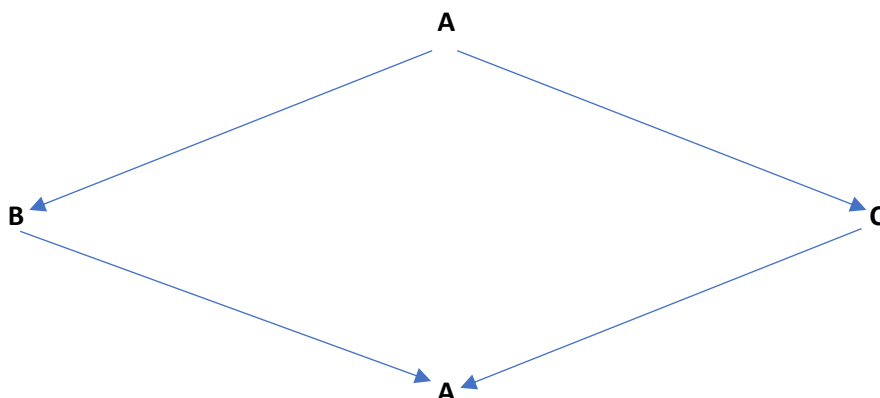
- ➔ From the above two tables no multivalued functional dependency is exist.
- ➔ Because each two consisting less than 3 attributes hence multivalued functional dependency is not possible with two attributes.

5th NORMAL FORM:

- ➔ A table is said to be in 5th Normal form it should be satisfied the below tow conditions
 - 1) It should be in 4th normal form table
 - 2) It should not have any Join Dependency

Join Dependency:

Def: if a table is decomposed into smaller tables and that leads to some loss of information or some additional information can be created then we should not decompose the table because that will lead to incorrect information.



Clarification:

- ➔ Suppose if you want to create a software with database connection there are some business logics and some data collected in a table.
- ➔ If you focus only on the data then we just have to see that if decompose the table into smaller tables which when combine leads to extra row of the data or less row of data then we stick to the original data as per 5NF.
- ➔ But if dividing table does not lead to any loss of information and using decompose relations, we can still verify all the facts about data then we should breakdown the relation into smaller relation.
- ➔ This is 5th normal form.